

ALIREZA SHIRI

Tehran, Iran

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SUMMARY

B.Sc. in Physics (Sharif University) and M.Sc. in AI & Robotics (University of Tehran). My physics background provides a robust foundation in data-driven concepts and a deep understanding of the mathematical principles behind AI algorithms. A fast, independent learner whose technical skills are rooted in advanced university coursework, I am eager to apply my expertise in AI, ML, and computational neuroscience to real-world challenges.

EDUCATION

M.Sc. in Computer Engineering – AI and Robotics 2024 - Present
University of Tehran, Tehran, Iran | Overall Grade: A+
Thesis: Alignment of Representations in Brain and Machine

B.Sc. in Physics 2018 - 2022
Sharif University of Technology, Tehran, Iran | Overall Grade: B+

MASTER'S THESIS RESEARCH

Alignment of Semantic Representations in Brains and Machines University of Tehran

- Objective & Encoding:** Designed a framework to align semantic representations across the human brain (EEG/fMRI) and AI (ResNet, BERT) using regression models validated via zero-shot generalization.
- Geometric Alignment:** Proved that projecting AI embeddings onto dominant principal components triggers macro-categorical clustering that aligns with human cognitive taxonomies.
- Ongoing Work:** Developing a Multimodal Contrastive Learning framework by training MLP projection layers on frozen modality encoders to map and compare representations in a shared similarity space.

SELECTED PROJECTS

For full source code and additional projects, please visit my GitHub profile.

Trustworthy ML & XAI: Implemented the FAAP algorithm for bias reduction, a teacher-student backdoor defense, evaluated PGD adversarial robustness, and applied global/local XAI (SHAP, LIME, Grad-CAM).

NLP to LLMs: Built comprehensive pipelines encompassing text classification, neural embeddings, LLM instruction-tuning, a Legal RAG system, and an intelligent TravelBot.

Computational Neuroscience: Applied custom spike sorting, time-resolved SVM decoding, Representational Dissimilarity Matrices (RDM), and Phase-Amplitude Coupling (PAC) for visual stimuli analysis.

CNN Image Classification: Developed deep learning pipelines using CNNs and transfer learning for accurate classification of medical skin lesions and agricultural plant diseases.

SKILLS

Programming & Tools	Python, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, $\text{\LaTeX}$
Familiar Tools	Linux, Docker, Git, FastAPI, n8n, Cursor, MIFlow, C#, .NET, SQL
NLP & Generative AI	Hugging Face, LangChain, LangGraph, LanceDB, LLMs, RAG
Advanced AI	Model Robustness, Domain Adaptation/Generalization, Contrastive Learning
Neuroscience & XAI	Brain Signal Processing, RSA, Grad-CAM, SHAP, LIME
Languages	Persian (Native), English (Intermediate / Actively improving)

HONORS & ACHIEVEMENTS

Ranked among the top 1% of participants nationwide in the University Entrance Exam for the Master's program.
Ranked among the top 1% of participants nationwide in the University Entrance Exam for the Bachelor's program.
Military service obligation fulfilled.